

PROMPT FOR DETERMINING THE 11-DIMENSIONAL KISSING NUMBER

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ABSTRACT. This document contains a full frontier-model research prompt for determining the exact kissing number in dimension 11. The current certified range is $604 \leq \tau_{11} \leq 868$; a complete resolution must match an exact spherical-code construction with a rigorous universal upper bound. Status snapshot: 20 July 2026.

1. Prompt

Current task statement

The kissing number τ_d is the maximum number of nonoverlapping unit spheres in $\mathbb{R}^d$ that can simultaneously touch a fixed unit sphere. Equivalently, τ_d is the maximum size of a set of unit vectors $x_1, \dots, x_N \in \mathbb{R}^d$ satisfying

$$\langle x_i, x_j \rangle \leq \frac{1}{2} \quad (i \neq j).$$

Determine τ_{11} exactly. As of the status date for this prompt, the rigorous range is

$$604 \leq \tau_{11} \leq 868.$$

Assume for purposes of this task that a complete resolution is reachable. Do not treat a new record construction or a modest upper-bound improvement as a complete answer.

Complete-resolution standard

A complete proof that $\tau_{11} = N$ must establish both:

- a lower bound by constructing N unit vectors in $\mathbb{R}^{11}$ with every pairwise inner product at most $1/2$;
- an upper bound proving that no configuration of $N+1$ unit vectors in $\mathbb{R}^{11}$ can satisfy the same inequalities;
- exact or rigorously interval-certified coordinates, Gram matrices, and optimization certificates, with no unquantified floating-point assumptions;
- a proof that every rank, positivity, representation-theoretic, or symmetry assumption used in an upper bound applies to arbitrary configurations.

What does not count

Partial progress is insufficient unless it logically implies the exact resolution above. In particular, the following do not count:

- improving the lower bound above 604 without a matching upper bound, or improving the upper bound below 868 without a matching construction;
- a numerical point cloud whose norms, pairwise angles, or rank are only approximately valid;
- a positive semidefinite Gram matrix of rank greater than 11;
- an SDP optimum reported in decimal form without a rigorous rational or interval certificate;
- an upper bound restricted to antipodal, lattice, code-derived, or symmetric configurations;
- a configuration for a different minimum angle, sphere radius convention, or ambient dimension.

Use multiagent v2 aggressively and dynamically. You have up to 64 concurrent agents available. Do not use a fixed assignment such as "N agents for strategy X." Instead, manage the search using the following heuristics:

- Begin with a genuinely diverse portfolio of approaches. Agents should explore substantially different formulations, invariants, reductions, algebraic viewpoints, structural inductions, decompositions, optimization methods, exact computational searches, and formal-proof routes.
- Do not tell most agents the currently favored approach. Preserve independence during early rounds so that agents do not all converge to the same attractive but incomplete reduction.
- Maintain an explicit registry of approach families. Group agents by the mathematical mechanism they are using, not by superficial wording. If too many agents converge to one family, redirect some toward underexplored formulations.
- Do not allow one route to dominate merely because it gives elegant reductions. A route that ends at a missing lemma equivalent in strength to the original problem is not close to completion unless it supplies a genuinely new proof of that lemma.
- When an approach stalls at a theorem-strength missing lemma, mark that route as blocked. Reopen it only when an agent proposes a materially new mechanism, invariant, construction, or exact experiment.
- Keep several incompatible proof or construction routes alive through multiple rounds. Cross-pollinate ideas only after independent agents have developed them far enough to expose their real strengths and gaps.
- Use adversarial agents throughout. Every candidate theorem, construction, certificate, reduction, and program must be attacked independently. Assign separate agents to mathematical correctness, completeness of case analysis, numerical exactness, implementation soundness, and novelty checking.
- Require agents to return concrete lemmas, constructions, equations, counterexamples to proposed sublemmas, executable encodings, or formally stated proof obligations. Reject status reports, vague optimism, and claims that a global compatibility step is "routine."
- The root agent should repeatedly synthesize, challenge, redirect, and launch new rounds. Do not stop after the first wave fails. Convert promising patterns into exact conjectures and send those conjectures to independent proof and falsification teams.

Problem-specific search portfolio

In the initial independent rounds, ensure that the portfolio includes, but is not limited to, the following genuinely different families:

- layered constructions from lattices, binary or ternary codes, constant-weight codes, laminated packings, spherical designs, and unions of group orbits;
- exact Gram-matrix search using algebraic entries, association schemes, coherent configurations, and representation-theoretic block diagonalization;
- sign-flipping, orbit splitting, switching, perturbation, augmentation, and asymmetric deformation of known high-cardinality configurations;
- MILP, SAT modulo exact inner products, branch-and-bound, interval nonlinear optimization, evolutionary search, and differentiable energy methods followed by exact reconstruction;
- Delsarte linear programming, semidefinite programming, three-point and higher-point bounds, and cluster or cap decompositions;
- polynomial positivity and sums-of-squares certificates on $[-1, 1/2]$, converted to exact rational inequalities;
- geometric combinatorics of contact graphs, local caps, spherical Voronoi regions, and dimension-reduction inequalities;
- a formal pipeline that turns numerical SDP output into a small exact certificate checked by rational arithmetic and interval bounds.

Mandatory adversarial audit

Any surviving candidate must be checked against every item below by agents that did not originate the candidate:

- check all vectors have norm exactly 1 or provide rigorously bounded normalization that preserves the $1/2$ threshold;
- check every one of the $N(N-1)/2$ pairwise inner products in exact or outward-rounded interval

arithmetic;

- check that the Gram matrix is positive semidefinite and has rank at most 11;
- ensure repeated vectors, antipodal conventions, and sphere-radius scaling have not altered the cardinality claim;
- for an upper bound, verify every polynomial nonnegativity claim and every SDP dual constraint with a small independent checker;
- audit any symmetry reduction to ensure the upper bound still ranges over all configurations;
- reproduce the construction and certificate using independent software stacks and compare hashes and exact invariants.

Required final artifact

The final response must contain both a lower-bound object and an upper-bound certificate meeting at the same integer:

- exact coordinates or an exact Gram matrix for the N -point spherical code;
- a verifier for norms, inner products, positive semidefiniteness, and rank;
- the complete LP/SDP/polynomial/geometric upper-bound derivation in checkable exact form;
- proof certificates, interval logs, source code, software versions, and hashes;
- a human-readable argument connecting the finite artifacts to the conventional kissing-number definition.

Public search may be used for standard spherical-code theory, known dimension-11 constructions, and published LP/SDP upper bounds. Do not search for a claimed unpublished exact value or return only a new record configuration.

Do not return merely because current approaches fail or agents report theorem-strength gaps. Continue launching new rounds, reopening blocked approaches only when there is a genuinely new mechanism, and searching for fresh formulations.

Return only when a complete resolution has been found and survives adversarial audit. Do not return a reduction, a special case, an isolated missing lemma, an improved bound that leaves the target unresolved, a "best effort" literature review, or an explanation of why the problem is difficult.

Spend at least 8 hours of sustained search before even considering return or termination. If the execution environment imposes a hard stop before a complete resolution, do not fabricate success or elapsed work: preserve all machine-checkable artifacts and report only the strongest rigorously verified derivation together with the exact remaining gap.